

Strategic Delay of Product Introductions

Subir Bose, Mehman Ismayilli, Piercarlo Zanchettin

September 1, 2026

Introduction

We study a durable good monopolist's incentives for the timing of a product introduction:

- **Finite economic life:** A unit bought late delivers fewer periods of service than the same unit bought early. Willingness to pay for the *same physical good* falls mechanically over the product cycle.
- **Marginal cost of the product declines within the product cycle:** the monopolist can supply the *same* good more cheaply later than it could at launch.

The interaction

Both forces push in opposite directions. Late service is worth less to the buyer, but late production is cheaper for the monopolist. *Who should be served, and when?*

The tension we study

- **First-best benchmark:** Under complete information, how a *benevolent social planner* compensates the lost period of service against the saved production cost.
 - Delay is efficient only when the cost saving is large relative to the buyer's valuation.
- **Monopolist's incentive:** A monopolist faces the same trade-off under *imperfect information*, but also uses timing as a *screening device*.
 - By delaying the product introduction, the monopolist separates types.
- 'Over-delaying' occurs when the monopolist introduces the product to some consumer types in a later period or never relative to the first-best solution.
- **Research question:** How does a durable-good monopolist's incentive to time the introduction of a product with *finite economic life* and *decreasing marginal cost* depart from the socially optimal timing, under **commitment** and **non-commitment** regimes?

Commitment regime –perfectly committing to intertemporal price offers made in period one

Contribution from two inter-related stances: *intertemporal price discrimination* and *damaged goods*

1. **Intertemporal price discrimination** arises in equilibrium.

- This **contradicts** with findings of [Coase \(1972\)](#), [Stokey \(1979, 1981\)](#), [Gul et al. \(1986\)](#);
- Aligns with the findings of:
 - [Bulow \(1982\)](#) – committing by limiting future production capacity;
 - [Lee and Lee \(1998\)](#) if a new better and quality version of the product is available by R&D;
 - [Moorthy and Png \(1992\)](#) if quality level of the product can be decided endogenously
 - [McAfee and Wiseman \(2008\)](#) – committing by how much to invest;
 - [Besbes and Lobel \(2015\)](#) if new customers enter the market in later periods, making demand stationary across periods with heterogenous consumers;
 - [Rochet and Thanassoulis \(2019\)](#) derive sufficiency conditions under Spence-Mirrless sorting conditions are failing.

¹All hypens are generated by Mehman not by AI

2. Damaged Goods

- Our monopolist's second period good can be thought as damaged good which also happens to be *perishable*
- Deneckere and McAfee (1996); McAfee (2007); Anderson and Dana (2009) assume that the damaged goods' marginal cost is not necessarily lower than original version, contrasting to our story. For this reason, they assume *increasing percentage difference condition/log supermodularity* for consumer valuations.
- In our model, we do not impose such condition on consumer valuations.

Literature Review – Non-commitment Regime

- Coase (1972); Stokey (1979, 1981); Gul et al. (1986) argue dynamic pricing is sub-optimal in continuous time horizon as time interval is small.
- Bagnoli et al. (1989) With discrete types and complete information the conjecture fails: the firm extracts rents above the marginal type. But it needs an infinite horizon because with a finite horizon the firm rushes to clear the market and the delay disappears.
- Ortner (2017). With *stochastic* marginal costs following a Brownian motion with drift, inefficient delay of the H-type arises: the firm waits for high-cost states, where serving the L-type is unprofitable, so the H-type stops waiting for a discount.

Our contribution here

Same qualitative distortion, i.e., over-delay of the H-type with a *two-period, deterministic* model, and driven by future cost *efficiency* rather than by high future costs.

Model

- A two-period model of a perfectly durable good market served by a monopolist, $t = \{1, 2\}$.
- A perfectly durable good with finite economic life, ends after period 2.
- Production technology evolves within the product's economic life such that

$$c_1 > c_2 \geq 0$$

Consumers:

- Differentiated according to their willing to pay for quality, V
- Low-type: V_L with relative mass size of μ
- High-type: V_H with relative mass size of $1 - \mu$
- Consumers know their own type and are forward-looking.
- Unit demand for the good, and a consumer's net flow of utility is

$$U_i = V_{t,i} - P_{t,i} \quad \text{for any } i = \{l, H\}$$

Timing of purchase matters

$\underbrace{2V_J - p_1}$
buy at launch: two periods of service

versus

$\underbrace{V_J - p_2}$
buy late: one period of service

Willingness to pay is $2V_J$ in period 1 and V_J in period 2.

- Technology and the distribution of types are common knowledge; the monopolist cannot observe an individual consumer's type.

Timing

- Delivers the product on time, receives a revenue equal to contractual price, and incurs production cost
 - no contracting cost
- Zero profit if nobody purchases the product
- **Objective:** To find and implement the consumption pattern that yields the highest profit in equilibrium.

First-best solution

Benevolent Social Planner with perfect information about consumer types is to optimize aggregate welfare.

For a generic type J , three distinct ‘production-consumption paths’ are available:

Path	Description	Net surplus
J_{12}	serve at launch, two periods of service	$2V_J - c_1$
J_2	serve late, one period of service, cheaper	$V_J - c_2$
J_0	never serve	0

- Delaying the product introduction buys a cost saving $c_1 - c_2$ at the price of one period of service worth V_J .
- So delay is efficient exactly when $V_J < c_1 - c_2$, provided the type is worth serving at all, $V_J > c_2$.

The planner's choice for a generic type

Lemma

The first-best production-consumption pattern of type J is

$$\begin{aligned} J_{12} & \text{ if } 2V_J - c_1 > \max\{V_J - c_2, 0\} \\ J_2 & \text{ if } V_J - c_2 > \max\{2V_J - c_1, 0\} \\ J_0 & \text{ if } 0 > \max\{2V_J - c_1, V_J - c_2\}. \end{aligned}$$

- J_2 beats $J_{12} \iff V_J - c_2 > 2V_J - c_1 \iff V_J < c_1 - c_2$.
- J_2 beats $J_0 \iff V_J > c_2$.
- Hence delay is ever optimal only if $c_2 < V_J < c_1 - c_2$, which requires the *significant cost saving* condition

$$c_2 < c_1 - c_2. \tag{A}$$

Proposition 1(A): first best when cost savings are significant

Proposition (First best, case $c_2 < c_1 - c_2$)

Under (A) the thresholds order as $c_1 - c_2 > \frac{c_1}{2} > c_2$ and

$$J_{12} \text{ if } V_J > c_1 - c_2, \quad J_2 \text{ if } c_1 - c_2 > V_J > c_2, \quad J_0 \text{ if } c_2 > V_J.$$

Applying this to both types (recall $V_H > V_L$):

	Condition	First best
(A1)	$c_2 < c_1 - c_2 < V_L < V_H$	both served at launch
(A2)	$c_2 < V_L < c_1 - c_2 < V_H$	H at launch, L late
(A3)	$V_L < c_2 < c_1 - c_2 < V_H$	H at launch, L never
(A4)	$c_2 < V_L < V_H < c_1 - c_2$	both served late
(A5)	$V_L < c_2 < V_H < c_1 - c_2$	H late, L never
(A6)	$V_L < V_H < c_2 < c_1 - c_2$	nobody served

(A1) and (A2) are the two benchmarks the rest of the talk uses.

Proposition 1(B): first best when cost savings are insignificant

Proposition (First best, case $c_2 > c_1 - c_2$)

Under (B), we have $c_2 > \frac{c_1}{2} > c_1 - c_2$, so J_2 is never optimal and

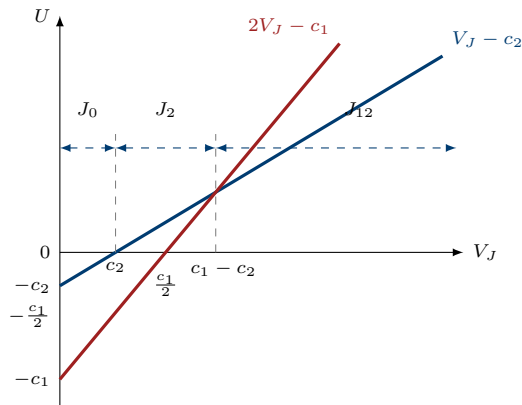
$$J_{12} : \quad \text{if } V_J > \frac{c_1}{2}; \quad J_0 : \quad \text{if } V_J < \frac{c_1}{2}.$$

	Condition	First best
(B1)	$\frac{c_1}{2} < V_L < V_H$	both served at launch
(B2)	$V_L < \frac{c_1}{2} < V_H$	H at launch, L never
(B3)	$V_L < V_H < \frac{c_1}{2}$	nobody served

Reading

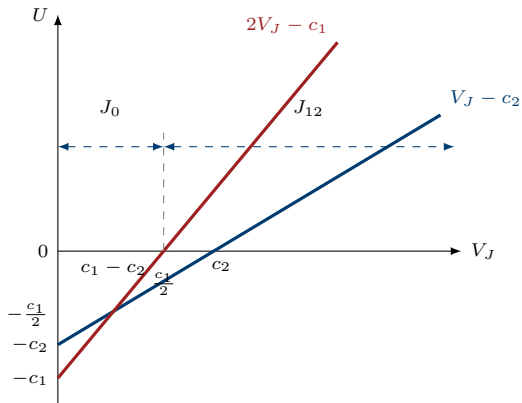
When the cost saving is small, *waiting has no social value at all*: the planner serves immediately or never.

First best when $c_1 - c_2 > c_2$



Significant cost savings: an intermediate region $c_2 < V_J < c_1 - c_2$ appears in which the planner *efficiently* delays type J to period 2.

First best when $c_1 - c_2 < c_2$



Insignificant cost savings: the J_2 region vanishes: the planner serves immediately or never.

What we will look for

Definition: over-delay

Type J 's consumption is **over-delayed** if postponing J to period 2 is optimal for the *monopolist* but sub-optimal for the *planner*.

- We do not characterise the optimality region of every consumption pattern. We ask only: *does the monopolist's optimum delay a type relative to Proposition 1?*

Monopolist's problem

What commitment means here

- The monopolist announces $\{p_1, p_2\}$ at the start of period 1.
- **First notion:** it cannot *renege* on an announced second-period price. A promised discount will be honoured.
- **Second notion:** it commits *not to offer any new contract* in period 2. A promise *not* to discount is also credible.

Why the second notion matters

It is what makes (H_{12}, L_0) , i.e., serve the H-type at full price, never serve the L-type, implementable. Without it, the H-type anticipates a period-2 discount to the L-type and refuses to pay $2V_H$.

The nine consumption patterns

	H-TYPE SERVED		
	at launch	only late	never
L NEVER	(H_{12}, L_0)	(H_2, L_0)	(H_0, L_0)
L AT LAUNCH	(H_{12}, L_{12})	(H_2, L_{12})	(H_0, L_{12})
L ONLY LATE	(H_{12}, L_2)	(H_2, L_2)	(H_0, L_2)

- **Column 2** is where the H-type's consumption is delayed.
- **Row 3** is where the L-type's consumption is delayed.
- (H_0, L_{12}) and (H_0, L_2) are **not implementable**: any price the L-type accepts, the H-type accepts too, since $V_H > V_L$.
- (H_0, L_0) needs $p_1 > 2V_H$, $p_2 > V_H$, and yields zero profit.

Pattern (H_{12}, L_0) : exclude the L-type

$$\max_{p_1} (1 - \mu)(p_1 - c_1)$$

subject to

$$2V_H - p_1 \geq V_H - p_2 \quad (IC_H)$$

$$2V_H - p_1 \geq 0 \quad (IR_H)$$

$$0 > 2V_L - p_1 \quad (IC_L^1)$$

$$0 > V_L - p_2 \quad (IC_L^2)$$

- Commitment makes the L-type's constraints slack; their IR 's bind trivially at $0 \geq 0$.
- IR_H binds — all surplus from two periods of H-consumption is extracted; IC_H is slack.

$$\boxed{p_1 = 2V_H, \quad p_2 \geq V_H, \quad \Pi_{(H_{12}, L_0)} = (1 - \mu)(2V_H - c_1).}$$

Pattern (H_{12}, L_{12}) : serve everybody at launch

$$\max_{p_1} (1 - \mu)(p_1 - c_1) + \mu(p_1 - c_1)$$

subject to

$$2V_H - p_1 \geq V_H - p_2 \quad (IC_H)$$

$$2V_H - p_1 \geq 0 \quad (IR_H)$$

$$2V_L - p_1 > V_L - p_2 \quad (IC_L)$$

$$2V_L - p_1 \geq 0 \quad (IR_L)$$

- A single uniform price must be acceptable to the L-type: IR_L binds, $p_1 = 2V_L$.
- Substituting: $p_2 \geq 2V_L - V_H$ and $p_2 \geq V_L$. Since $V_H > V_L$, the binding requirement is $p_2 \geq V_L$.

$$\boxed{p_1 = 2V_L, \quad p_2 \geq V_L, \quad \Pi_{(H_{12}, L_{12})} = 2V_L - c_1.}$$

Pattern (H_{12}, L_2) : intertemporal price discrimination

$$\max_{p_1, p_2} (1 - \mu)(p_1 - c_1) + \mu(p_2 - c_2)$$

subject to

$$2V_H - p_1 \geq V_H - p_2 \quad (IC_H)$$

$$2V_H - p_1 \geq 0 \quad (IR_H)$$

$$2V_L - p_1 < V_L - p_2 \quad (IC_L)$$

$$V_L - p_2 \geq 0 \quad (IR_L)$$

In equilibrium IR_L and IC_H bind:

$$V_L - p_2 = 0, \quad 2V_H - p_1 = V_H - p_2 \implies \boxed{p_1 = V_H + V_L, \quad p_2 = V_L.}$$

$$\Pi_{(H_{12}, L_2)} = \mu(V_L - c_2) + (1 - \mu)(V_H + V_L - c_1).$$

Information rent: the H-type keeps $V_H - V_L > 0$ and is exactly indifferent between buying early and buying late.

Pattern (H_2, L_0) : serve nobody at launch

$$\max_{p_2} (1 - \mu)(p_2 - c_2)$$

subject to

$$V_H - p_2 \geq 2V_H - p_1 \quad (IC_H)$$

$$V_H - p_2 \geq 0 \quad (IR_H)$$

$$0 > 2V_L - p_1 \quad (IC_L^1)$$

$$0 > V_L - p_2 \quad (IC_L^2)$$

- IR_H binds: the whole of the H-type's *one-period* surplus is extracted.
- Period-1 price is set prohibitively high.

$$p_1 \geq 2V_H, \quad p_2 = V_H, \quad \Pi_{(H_2, L_0)} = (1 - \mu)(V_H - c_2).$$

Pattern (H_2, L_{12}) : the low type first

$$\max_{p_1, p_2} (1 - \mu)(p_2 - c_2) + \mu(p_1 - c_1)$$

subject to

$$V_H - p_2 \geq 2V_H - p_1 \quad (IC_H)$$

$$V_H - p_2 \geq 0 \quad (IR_H)$$

$$2V_L - p_1 \geq V_L - p_2 \quad (IC_L)$$

$$2V_L - p_1 \geq 0 \quad (IR_L)$$

IR_L and IC_H bind, giving $p_1 = 2V_L$ and $p_2 = \max\{2V_L - V_H, 0\}$:

$$\Pi_{(H_2, L_{12})} = \mu(2V_L - c_1) + (1 - \mu)(\max\{2V_L - V_H, 0\} - c_2).$$

Note

This pattern can *never* be first best: if serving the L-type at launch is worthwhile, serving the H-type at launch is worthwhile too, since $V_H > V_L$.

Pattern (H_2, L_2) : serve everybody, late

$$\max_{p_2} (1 - \mu)(p_2 - c_2) + \mu(p_2 - c_2)$$

subject to

$$V_H - p_2 \geq 2V_H - p_1 \quad (IC_H)$$

$$V_H - p_2 \geq 0 \quad (IR_H)$$

$$V_L - p_2 \geq 2V_L - p_1 \quad (IC_L)$$

$$V_L - p_2 \geq 0 \quad (IR_L)$$

IR_L binds at $p_2 = V_L$; substituting into the IC 's gives $p_1 \geq V_H + V_L$ and $p_1 \geq V_L$, so $p_1 \geq V_H + V_L$ is the binding one.

$p_1 \geq V_H + V_L, \quad p_2 = V_L, \quad \Pi_{(H_2, L_2)} = V_L - c_2.$
--

Profits at optimal prices: the full menu

Pattern	p_1	p_2	Profit
(H_{12}, L_0)	$2V_H$	$\geq V_H$	$(1 - \mu)(2V_H - c_1)$
(H_{12}, L_{12})	$2V_L$	$\geq V_L$	$2V_L - c_1$
(H_{12}, L_2)	$V_H + V_L$	V_L	$\mu(V_L - c_2) + (1 - \mu)(V_H + V_L - c_1)$
(H_2, L_0)	$\geq 2V_H$	V_H	$(1 - \mu)(V_H - c_2)$
(H_2, L_{12})	$2V_L$	$\max\{2V_L - V_H, 0\}$	$\mu(2V_L - c_1) + (1 - \mu)(\max\{2V_L - V_H, 0\} - c_2)$
(H_2, L_2)	$\geq V_H + V_L$	V_L	$V_L - c_2$
(H_0, L_0)	$> 2V_H$	$> V_H$	0

Everything that follows is a comparison inside this table.

Proposition 2: over-delaying the H-type is impossible

Proposition (Impossibility of over-delaying the H-type)

Under full commitment, over-delay (relative to the first best) of the H-type's consumption can never arise in equilibrium.

Strategy of proof

Over-delay of the H-type requires that some *column-2* pattern be optimal while the planner would *not* delay the H-type. We show that each column-2 pattern beats its column-1 counterpart *only if*

$$c_1 - c_2 > V_H > c_2, \quad (C_{H_2})$$

which is exactly the first-best condition under which the planner *does* delay the H-type.

Proof of Proposition 2 (i): patterns with L_0 and L_2

(H_2, L_0) **versus** (H_{12}, L_0) . Optimality requires

$$\Pi_{(H_2, L_0)} > 0 \implies V_H > c_2,$$

$$\Pi_{(H_2, L_0)} > \Pi_{(H_{12}, L_0)} \implies V_H - c_2 > 2V_H - c_1 \implies c_1 - c_2 > V_H.$$

(H_2, L_2) **versus** (H_{12}, L_2) .

$$\Pi_{(H_2, L_2)} > 0 \implies V_L > c_2 \implies V_H > c_2,$$

$$\Pi_{(H_2, L_2)} > \Pi_{(H_{12}, L_2)} \implies V_L - c_2 > V_L + V_H - c_1 \implies c_1 - c_2 > V_H.$$

Both give $c_1 - c_2 > V_H > c_2$, i.e. condition (C_{H_2}) .

Proof of Proposition 2 (ii): the pattern (H_2, L_{12})

$$\Pi_{(H_2, L_{12})} = \mu(2V_L - c_1) + (1 - \mu)(\max\{2V_L - V_H, 0\} - c_2).$$

- If $2V_L - V_H < c_2$ this can never be the global maximum:
 - if also $2V_L - c_1 < 0$, inactivity yields more (zero);
 - if $2V_L - c_1 > 0$, then (H_{12}, L_{12}) yields more.

So a necessary condition is $2V_L - V_H > c_2$, and since $V_H > 2V_L - V_H$ this implies $V_H > c_2$.

- Next, (H_2, L_{12}) must beat (H_{12}, L_{12}) :

$$2V_L - V_H - c_2 > 2V_L - c_1 \implies c_1 - c_2 > V_H.$$

Again (C_{H_2}) . ■

Corollary 1

The monopolist never delays the H-type to period 2 when $V_H > c_1 - c_2 > c_2$.

(A_{1H})

Why the impossibility is intuitive

- Delaying the H-type costs the monopolist the H-type's *second* period of service, worth V_H in revenue, and buys the cost saving $c_1 - c_2$.
- That is *exactly* the planner's trade-off for the H-type: the monopolist extracts the H-type's full surplus in both (H_{12}, L_0) and (H_2, L_0) , so private and social incentives coincide on the margin.
- Screening gives no extra reason to delay the *top* type. There is nobody above the H-type whose incentive constraint delay could relax.

This is exactly what will break without commitment

Under non-commitment the monopolist loses (H_{12}, L_0) from its menu. The comparison above is then no longer available, and delaying the H-type becomes a substitute for a promise it can no longer make.

Proposition 3: over-delaying the L-type

Proposition (Intertemporal price discrimination)

Over-delay (relative to the first best) of the L-type's consumption characterises the monopoly equilibrium if and only if

$$V_H > V_L > c_1 - c_2 > c_2, \quad (\text{A1})$$

$$\frac{V_H - V_L}{V_H - c_2} < \mu < \frac{V_H - V_L}{V_H - (c_1 - c_2)}. \quad (\text{C1})$$

- (A1): both valuations high $\Rightarrow$ the planner serves *both* types at launch; *any* delay is over-delay.
- (C1): the L-type's mass is large enough — but not too large.

Proof of Proposition 3 (i): sufficiency

Under (A1), Corollary 1 rules out all of column 2, and $(H_0, \cdot)$ is either not implementable or dominated. Three candidates remain:

$$\begin{aligned}\Pi_{(H_{12}, L_{12})} &= 2V_L - c_1, & \Pi_{(H_{12}, L_0)} &= (1 - \mu)(2V_H - c_1), \\ \Pi_{(H_{12}, L_2)} &= \mu(V_L - c_2) + (1 - \mu)(V_H + V_L - c_1).\end{aligned}$$

(H_{12}, L_2) is optimal *only if*

$$\begin{aligned}\Pi_{(H_{12}, L_2)} > \Pi_{(H_{12}, L_{12})} &\implies \mu < \frac{V_H - V_L}{V_H - (c_1 - c_2)}, \\ \Pi_{(H_{12}, L_2)} > \Pi_{(H_{12}, L_0)} &\implies \mu > \frac{V_H - V_L}{V_H - c_2}.\end{aligned}$$

For the interval to be non-empty we need

$$V_H - c_2 > V_H - (c_1 - c_2) \iff c_2 < c_1 - c_2,$$

i.e. condition (A), guaranteed by (A1). **Significant cost savings are necessary for the interval to exist at all.**

Proof of Proposition 3 (ii): necessity

Suppose the L-type's consumption is over-delayed. Then:

(i) The planner must serve the L-type at launch, so by Lemma 1

$$V_H > V_L > \max\left\{c_1 - c_2, \frac{c_1}{2}\right\}. \quad (A1_A)$$

This holds in one of two ways: (A1) if $\frac{c_1}{2} > c_2$, or (B1) if $\frac{c_1}{2} < c_2$.

(ii) A row-3 pattern must be implemented. (H_0, L_2) is not implementable, so it is (H_2, L_2) or (H_{12}, L_2) .

- Under (A1), Corollary 1 makes (H_2, L_2) profit-dominated by (H_{12}, L_2) .
- Under (B1), $c_2 > \frac{c_1}{2}$ and $\Pi_{(H_{12}, L_{12})} = 2(V_L - \frac{c_1}{2}) > V_L - c_2 = \Pi_{(H_2, L_2)}$.

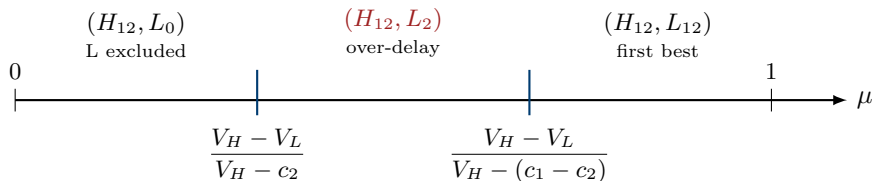
So (H_{12}, L_2) must be the optimum.

But (B1) implies $c_1 - c_2 < c_2$, which violates the necessary condition (A). Hence only (A1) survives, and conditional on (A1), (C1) is necessary and sufficient. ■

Reading condition (C1): why the L-type's mass matters

Under commitment the low future price is *announced in advance*. It creates a third option for the H-type: *wait*.

- To stop the H-type waiting, the monopolist must leave them a rent of $V_H - V_L$. Foregone revenue is proportional to $1 - \mu$.
- The compensation is second-period profit from the L-type, proportional to μ .



- μ too small: not worth the rent, so exclude the L-type entirely.
- μ too large: uniform pricing at launch dominates, so coinciding with the first best.

Welfare under commitment

Under (A1) the first best is (H_{12}, L_{12}) :

$$W_{(H_{12}, L_{12})} = (1 - \mu)(2V_H - c_1) + \mu(2V_L - c_1).$$

The monopolist implements (H_{12}, L_2) :

$$W_{(H_{12}, L_2)} = (1 - \mu)(2V_H - c_1) + \mu(V_L - c_2).$$

Welfare loss from intertemporal price discrimination

$$\Delta W = \mu(V_L - (c_1 - c_2)) > 0 \quad \text{under (A1).}$$

- Compare with the alternative distortion, complete exclusion (H_{12}, L_0) , which costs $\mu(2V_L - c_1)$, strictly larger under (A1).
- **Delay is the moderate distortion.** The monopolist's screening instrument destroys less surplus than outright exclusion would.

Relaxing commitment

- The monopolist may now offer a *new* contract in period 2 after the period-1 contract has been announced.
- **The first best is unchanged.** Only the monopolist's feasible set moves.
- The set of implementable patterns does not *grow*. It *shrinks* by exactly one element.

(H_{12}, L_0) is no longer implementable

Once the H-type has bought at $p_1 = 2V_H$, the monopolist has a strict incentive to offer $p_2 = V_L$ to the remaining L-types whenever $V_L > c_2$. The H-type anticipates this and refuses to pay $2V_H$.

$$(H_{12}, L_0) \longrightarrow (H_{12}, L_2).$$

What survives, and why it matters

Still valid from the commitment analysis:

$$\Pi_{(H_{12}, L_{12})} > \Pi_{(H_2, L_{12})}, \quad \Pi_{(H_{12}, L_2)} > \Pi_{(H_2, L_2)}.$$

No longer available:

$$\Pi_{(H_{12}, L_0)} > \Pi_{(H_2, L_0)} \quad \text{because the left-hand pattern cannot be implemented.}$$

Consequence

Corollary 1 no longer rules out column 2. (H_2, L_0) becomes a live candidate, and it is the *only* remaining way to serve the H-type without a subsequent discount to the L-type because after period 2 there *is* no subsequent period.

Under (A2), $c_2 < V_L < c_1 - c_2 < V_H$, the relevant comparison is therefore

$$(H_2, L_0) \quad \text{versus} \quad (H_{12}, L_2) \quad \text{versus} \quad (H_{12}, L_{12}).$$

Proposition 4: over-delaying the H-type

Proposition (Over-delay of the H-type without commitment)

Over-delay of the H-type's consumption by one period characterises the monopoly equilibrium if and only if

$$V_H > c_1 - c_2 > V_L > c_2, \quad (\text{A2})$$

$$\mu < \frac{c_1 - c_2 - V_L}{c_1 - 2c_2}. \quad (\text{C2})$$

- (A2): the planner serves H at launch and L late.
- (C2): the L-type's mass is *small*. Note $c_1 > 2c_2$ and $c_1 - c_2 > V_L$ under (A2), so the bound is strictly positive.
- Equilibrium pattern: (H_2, L_0) , meaning *nobody* at launch, only the H-type late.

Proof of Proposition 4 (i): the limit $\mu \rightarrow 0$

$$\begin{aligned}\Pi_{(H_2, L_0)} &= (1 - \mu)(V_H - c_2), & \Pi_{(H_{12}, L_{12})} &= 2V_L - c_1, \\ \Pi_{(H_{12}, L_2)} &= \mu(V_L - c_2) + (1 - \mu)(V_L + V_H - c_1).\end{aligned}$$

Taking $\mu \rightarrow 0$:

$$\begin{aligned}\Pi_{(H_2, L_0)} > \Pi_{(H_{12}, L_2)} &\implies V_H - c_2 > V_L + V_H - c_1 \implies c_1 - c_2 > V_L, \\ \Pi_{(H_2, L_0)} > \Pi_{(H_{12}, L_{12})} &\implies V_H - c_2 > 2V_L - c_1 \implies c_1 - c_2 > 2V_L - V_H.\end{aligned}$$

Since $V_H > V_L$, $\max\{V_L, 2V_L - V_H\} = V_L$, so the binding requirement is

$$c_1 - c_2 > V_L.$$

Combined with (A_{1H}) , $V_H > c_1 - c_2 > c_2$, this gives either (A2) or

$$V_H > c_1 - c_2 > c_2 > V_L. \tag{A3}$$

Proof of Proposition 4 (ii): ruling out (A3)

- Under **(A3)**, $V_L < c_2$: serving the L-type is *unprofitable in every period*.
- The H-type therefore knows no discount will ever come — the monopolist's promise is credible *because it is self-enforcing*.
- So (H_{12}, L_0) is implementable even without commitment, and matches the first best. **No over-delay.**
- Under **(A2)**, $V_L > c_2$: serving the L-type late is profitable, the promise is not credible, and (H_{12}, L_0) collapses.
- Only (H_2, L_0) and (H_{12}, L_2) remain, and only the former involves over-delay.

Proof of Proposition 4 (iii): condition (C2)

Compare the two survivors for arbitrary μ :

$$\begin{aligned}\Pi_{(H_2, L_0)} &> \Pi_{(H_{12}, L_2)} \\ (1 - \mu)(V_H - c_2) &> \mu(V_L - c_2) + (1 - \mu)(V_L + V_H - c_1) \\ (1 - \mu)(V_H - c_2 - V_L - V_H + c_1) &> \mu(V_L - c_2) \\ (1 - \mu)(c_1 - c_2 - V_L) &> \mu(V_L - c_2) \\ c_1 - c_2 - V_L &> \mu(c_1 - 2c_2) \\ \mu &< \frac{c_1 - c_2 - V_L}{c_1 - 2c_2}.\end{aligned}\tag{C2}$$

Also, under (A2), $\Pi_{(H_{12}, L_2)} > \Pi_{(H_{12}, L_{12})}$ for *every* μ , since $\mu(V_H - (c_1 - c_2)) < V_H - V_L$ whenever $V_L < c_1 - c_2$. ■

The economics of waiting

- With commitment the monopolist would choose (H_{12}, L_0) : extract $2V_H$ from the H-type and promise never to discount.
- Without commitment that promise is empty *as long as a period 2 exists in which the L-type can be served*.
- **The finite economic life is the key.** By waiting until period 2, the monopolist reaches a date after which no further contract is possible. The promise becomes credible by construction.

The price of credibility

$$\underbrace{V_H - V_L}_{\text{rent saved on the H-type}} \quad \text{versus} \quad \underbrace{V_H}_{\text{one period of H-service lost}} \quad - \quad \underbrace{(c_1 - c_2)}_{\text{cost saved}}$$

Waiting pays only when the cost saving $c_1 - c_2$ is large, and only when there are few L-types to sell to in the meantime.

Welfare under non-commitment

Under (A2) the first best is (H_{12}, L_2) :

$$W_{(H_{12}, L_2)} = (1 - \mu)(2V_H - c_1) + \mu(V_L - c_2).$$

The monopolist implements (H_2, L_0) :

$$W_{(H_2, L_0)} = (1 - \mu)(V_H - c_2).$$

Welfare loss: two components

$$\Delta W = \underbrace{(1 - \mu)(V_H - (c_1 - c_2))}_{\text{H-type served one period late}} + \underbrace{\mu(V_L - c_2)}_{\text{L-type excluded entirely}} > 0$$

Both terms are strictly positive under (A2).

Two distortions at once in contrast with the single, moderate distortion of the commitment regime.

The two regimes side by side

	COMMITMENT	NO COMMITMENT
Relevant first best	(A1): both at launch	(A2): H at launch, L late
Equilibrium pattern	(H_{12}, L_2)	(H_2, L_0)
Over-delay of H	impossible	possible
Over-delay of L	yes, one period	L excluded altogether
Condition on μ	interior interval (C1)	upper bound (C2), μ small
Cost savings needed	$c_2 < c_1 - c_2$	$c_2 < c_1 - c_2$, and $c_1 - c_2 > V_L$
Welfare loss	$\mu(V_L - (c_1 - c_2))$	$(1 - \mu)(V_H - (c_1 - c_2)) + \mu(V_L - c_2)$

- Relaxin commitment does not merely reduce profit but it changes *which* type is distorted.

Benchmark 1: constant marginal cost, $c_1 = c_2 = c$

Commitment regime.

- Delay buys no cost saving. The problem collapses to the classical durable-good monopoly.
- The firm serves either the H-type only, or both types, at launch depending on μ .
- Formally, $c_1 = c_2$ violates transitivity in (C1): the interval is empty. *Intertemporal price discrimination cannot arise.*

Non-commitment regime.

- If $V_L < c$, the planner never serves the L-type; the monopolist can implement (H_{12}, L_0) credibly and delay is strictly dominated.
- If $V_L > c$, (H_{12}, L_0) is not implementable; the firm chooses between (H_{12}, L_2) and (H_{12}, L_{12}) according to μ . The H-type is never delayed; the L-type may be.

Benchmark 2: small cost savings, $c_1 - c_2 < c_2$

- The higher effective second-period cost does not make waiting attractive enough.
- In the *commitment* regime, one of the necessary optimality conditions in Proposition 3, namely (A) is violated. The firm serves the H-type only, or both types, at launch.
- In the *non-commitment* regime, the cost advantage cannot compensate the forgone first-period revenue, so (H_2, L_0) is never optimal and behaviour reverts to the constant-cost benchmark.

Take-away

Both distortions we identify are phenomena of *fast-improving production technology*. Slow cost decline restores immediate-or-never service.

Relation to Ortner (2017), revisited

- Ortner obtains over-delay of the H-type with *stochastic* costs and an *infinite* horizon. The H-type buys in a high-cost state because costs and therefore prices are expected to remain high, so no lower price is offered to the L-type.
- In our setting, this translates into: ‘the firm waits for a state in which its marginal revenue from the L-type falls below the marginal cost. The promise not to discount then becomes self-enforcing.’
- That is the mechanism of our condition **(A3)**: when $V_L < c_2$, the H-type can be served from period 1 with no dynamic inconsistency.

The difference

Our over-delay does *not* require costs to be high in the future. It arises precisely because they will be *low*, and because the product *dies*. Finite economic life substitutes for persistence in costs.

Conclusion

- ① **Commitment.** If both valuations are high, the first best serves both types at launch. The monopolist can never over-delay the H-type, but *will* over-delay the L-type, practising intertemporal price discrimination, when future cost savings are large and the L-type's mass lies in an interior interval.
- ② **No commitment.** The monopolist can over-delay *both* types. Serving the H-type exclusively is dynamically consistent only in period 2, because the product's economic life expires afterwards. The L-type is then excluded altogether.
- ③ **The driver throughout** is the size of the future cost saving relative to valuations, not the shape of the valuation distribution. No increasing percentage differences, no log-supermodularity, no violation of Spence–Mirrlees.

Where this leaves us

What the model buys

A two-period, deterministic, discrete-type model delivers over-delay of the top type — previously obtained only with stochastic costs and an infinite horizon.

Open directions

- More than two periods: does the exclusion result survive when the firm has several chances to discount before the product dies?
- Continuous types with finite economic life.
- Endogenous economic life such as the firm choosing when to obsolete its own product.

Thank you.

References I

- Anderson, E. T. and Dana, J. D. (2009). When Is Price Discrimination Profitable? *Management Science*, 55(6):980–989.
- Bagnoli, M., Salant, S. W., and Swierzbinski, J. E. (1989). Durable-Goods Monopoly with Discrete Demand. *Journal of Political Economy*, 97(6):1459–1478.
- Besbes, O. and Lobel, I. (2015). Intertemporal price discrimination: Structure and computation of optimal policies. *Management Science*, 61(1):92–110.
- Bulow, J. I. (1982). Durable-goods Monopolists. *Journal of Political Economy*, 90(2):314–332.
- Coase, R. H. (1972). Durability and Monopoly. *The Journal of Law and Economics*, 15(1):143–149.
- Deneckere, R. J. and McAfee, R. P. (1996). Damaged Goods. *Journal of Economics & Management Strategy*, 5(2):149–174.
- Gul, F., Sonnenschein, H., and Wilson, R. (1986). Foundations of dynamic monopoly and the Coase conjecture. *Journal of Economic Theory*, 39(1):155–190.
- Lee, I. H. and Lee, J. (1998). A theory of economic obsolescence. *The Journal of Industrial Economics*, 46(3):383–401.
- McAfee, R. P. (2007). Pricing Damaged Goods. *Economics*, 1(1).
- McAfee, R. P. and Wiseman, T. (2008). Capacity choice counters the Coase conjecture. *The Review of Economic Studies*, 75(1):317–331.
- Moorthy, K. S. and Png, I. P. L. (1992). Market Segmentation, Cannibalization, and the Timing of Product Introductions. *Management Science*, 38(3):345–359.
- Ortner, J. (2017). Durable goods monopoly with stochastic costs: Durable goods monopoly. *Theoretical Economics*, 12(2):817–861.
- Rochet, J. and Thanassoulis, J. (2019). Intertemporal price discrimination with two products. *The RAND Journal of Economics*, 50(4):951–973.
- Stokey, N. L. (1979). Intertemporal price discrimination. *The Quarterly Journal of Economics*, pages 355–371.
- Stokey, N. L. (1981). Rational expectations and durable goods pricing. *The Bell Journal of Economics*, 12(1):112–128.